MOBILE DATA LEAKAGE SANDBOX
(under Prof. Mohanram)

IMT2022008 Harsh Dhruv
IMT2022081 Rudra Pathak
IMT2022103 Anurag Ramaswamy

Code link: https://github.com/hrdhruv/MobileSandbox

MobileSandbox PE Demo

Android app permission risk analysis with pre-install APK sandboxing

- Scans installed apps and selected APK/split APK files
- Scores privacy/security risk from 0-100
- Classifies apps as SAFE, HANDLE WITH CARE, SUSPICIOUS, or DANGEROUS
- Stores scan history and accepts Safe/Malware feedback

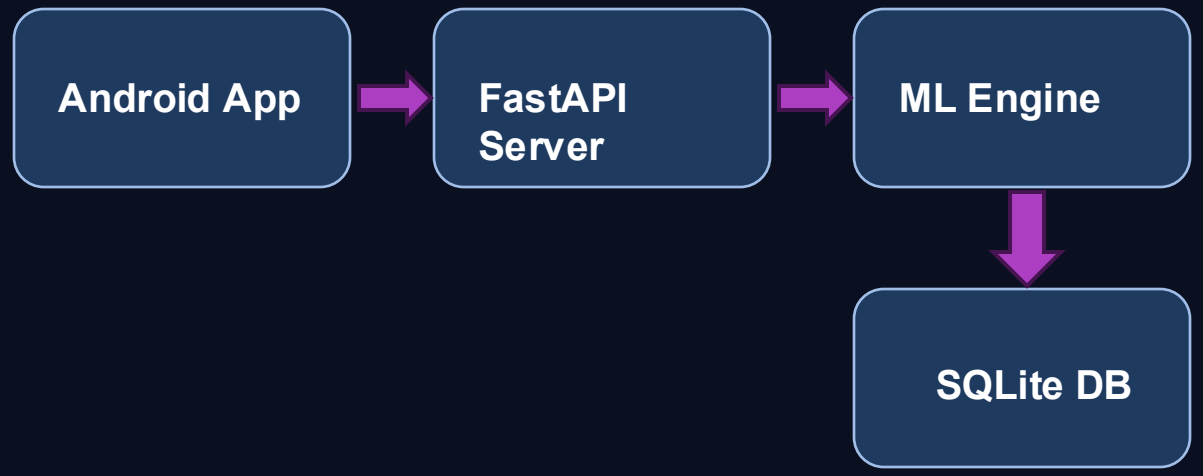
0 - 100

Risk Score
Permission + ML + Bayesian signals

Android Kotlin client + FastAPI ML analysis server + SQLite persistence

System Architecture & Data Flow

- Android app reads package name and permissions using PackageManager
- Pre-install APKs are copied into app-private cache before inspection
- FastAPI /analyze endpoint receives package_name + permissions
- ML engine returns score, risk level, PII list, sensitive list, threats, confidence
- SQLite stores scan history and user feedback



Endpoints: /analyze, /feedback, /scan/history, /stats

Hybrid Risk Scoring Workflow

- Signal A: static per-permission risk weights for SMS, contacts, location, camera, install packages, etc.
- Signal B: GradientBoostingClassifier probability signal from dataset/model files
- Signal C: safe-biased Bayesian prior so unknown apps do not start as high risk
- Final blend: 58% static + 24% ML + 18% prior when ML is available
- Calibration: dangerous combos, sparse-permission dampener, high-risk volume logic, known-safe caps, confidence score

SAFE	< 30
HANDLE WITH CARE	30-54
SUSPICIOUS	55-74
DANGEROUS	>= 75

Output: risk_level, score, PII list, sensitive list, detected threats, confidence

Application Workflow & Features

- Installed scan: tap scan, collect non-system apps, send each permission list to backend, show result cards
- Result card: risk badge, score meter, PII permissions, sensitive permissions, Safe/Malware feedback buttons
- Pre-install scan: user selects APK or split APK bundle, app extracts permissions without installing
- Decision step: install through Android PackageInstaller or reject and delete staged sandbox copy
- History tab: loads /scan/history, shows recent scans and total scan count

Installed Apps

- Scan current apps
- Score each app
- Submit feedback

Pre-install APK

- Stage in sandbox
- Analyze before install
- Install or reject

History

- SQLite records
- Recent scan list
- Duplicate guard